PicoLab

Hackathon Submission

A visual AI learning coach that turns STEM mistakes into learning signals.

PURPOSE

PicoLab helps students learn STEM problems through an interactive loop: problem intake, structured confirmation, step-by-step solving, visual explanation, personalized practice, and growth tracking.

It is built around a simple idea: mistakes should not be treated as failures. They should become learning signals that show students what to practice next.

PROBLEM

Many students get a numeric answer partially correct but still misunderstand the concept, formula, graph, or unit behind it. Traditional tools often give answers, but rarely diagnose the type of mistake or convert it into a clear practice path.

SOLUTION — A GUIDED WORKFLOW

1. Add or scan a STEM problem.
2. Confirm extracted values, units, target variable, and formulas.
3. Solve step-by-step in Smart Notebook.
4. Detect learning signals (unit mismatch, formula choice, graph reading, algebra).
5. Open a contextual Visual Lab explanation.
6. Practice via daily challenges and Pico's recommended missions.
7. Track signals in Growth Map and follow a personalized Roadmap.

KEY FEATURES

- **Add Problem —** type, scan, or enter formula-based STEM problems.
- **Scan & Confirm —** Pico structures known values, units, target, formula.
- **Smart Notebook —** step-by-step solving with contextual feedback.
- **Learning Signals —** mistakes classified by units, formulas, graphs, algebra, concepts, reading.
- **Visual Lab —** interactive visuals linked to the detected signal.
- **Practice Missions —** daily challenge, recommended practice, optional random missions.
- **Growth Map —** tracks recurring learning signals over time.
- **Roadmap —** converts signals into a personalized learning path.
- **Ask Pico —** contextual coaching drawer with mock/fallback behavior.

TECHNICAL IMPLEMENTATION

- React + Vite + TypeScript frontend.
- Tailwind-based design system.
- Express + TypeScript mock backend (REST).
- Backend-first API client with local fallback.
- Deterministic diagnostic engine for signals.
- Persistent progress via localStorage / sessionStorage.
- Provider-ready for future real AI integration.

DEMO FLOW

Problem -> Confirm -> Smart Notebook -> Learning Signal -> Visual Lab -> Practice Mission -> Growth Map -> Roadmap

PROJECT STATUS

PicoLab is a functional hackathon prototype. The demo uses deterministic mock/backend logic and a canonical physics problem to show the complete student learning loop. The architecture is ready for real AI providers, broader STEM domains, and richer visual simulations.